Project 2: Credit Analytics

(First discussion: Oct 14; Last questions: Oct 28; Deadline: Nov 4; In charge: Zhexin Wu)

This project is about credit analytics for consumer loans. The goal is to estimate risk profiles of individuals applying for a loan. For simplicity, we work with artificially generated data and only consider three borrower characteristics: age, monthly income and employment status. In reality, the availability of good data is important, and typically, many more features are taken into account.

Submission and report. Submit one fully executed Jupyter notebook in which all requested quantities are printed or shown in tables/plots. Briefly interpret your results where asked. At the end of the notebook, include a concise report in Markdown cells under the heading *Project Report*. The report should summarize your approach, model selection (here: your comparison of the classifiers and of the lending strategies), key results, interpretation, robustness, and limitations. Detailed computations and intermediate results should remain in the main body of the notebook.

Code quality. Keep your analysis readable and reproducible. Avoid unnecessary duplication and use small reusable helper functions where appropriate. Classes or object-oriented design are not required.

Grading. The implementation and results account for 80% of the project grade, and the Project Report accounts for 20%.

1. Let $m = 20000, n = 10000$ and simulate $m+n$ vectors $x_i = (x_{i1}, x_{i2}, x_{i3}) \in \mathbb{R}^3$, $i = 1, \dots, m+n$, with

   - x_{i1} = age in $[18, 80]$ (from the continuous uniform distribution)
   - x_{i2} = monthly income in CHF 1000 in $[1, 15]$ (from the continuous uniform distribution)
   - x_{i3} = salaried/self-employed in $\{0, 1\}$, where 0=salaried and 1=self-employed (probability of being self-employed is 10%)

   such that x_{i1}, x_{i2}, x_{i3} are independent.

   a) Compute the empirical means and standard deviations of x_{i1}, x_{i2} and x_{i3} over $i = 1, \dots, m$.

   b) Can you think of additional features (besides age, income, salaried/self-employed) that could be relevant in reality?

2. Let ξ_i, $i = 1, \dots, m+n$ be independent random variables that are uniformly distributed on $(0, 1)$ and $\psi \colon \mathbb{R} \to (0, 1)$ the logistic (or sigmoid) function given by

$$\psi(z) = \frac{e^z}{1+e^z} = \frac{1}{1+e^{-z}}.$$

   Consider two functions $p_1, p_2 \colon \mathbb{R}^3 \to (0, 1)$ of the form

$$\begin{aligned} p_1(x_i) &= \psi\left(13.3 - 0.33x_{i1} + 3.5x_{i2} - 3x_{i3}\right) \\ p_2(x_i) &= \psi\left(5 - 10\left[1_{(-\infty,25)}(x_{i1}) + 1_{(75,\infty)}(x_{i1})\right] + 1.1x_{i2} - x_{i3}\right) \end{aligned}$$

   and generate two artificial data sets $(x_i, y_i^{(1)})$ and $(x_i, y_i^{(2)})$, $i = 1, \dots, m+n$, by setting

$$y_i^{(1)} = \begin{cases} 1 & \text{if } \xi_i \le p_1(x_i), \\ 0 & \text{otherwise,} \end{cases} \quad \text{and} \quad y_i^{(2)} = \begin{cases} 1 & \text{if } \xi_i \le p_2(x_i), \\ 0 & \text{otherwise.} \end{cases}$$

(We use the convention that $y_i^{(s)} = 1$ is a good borrower whereas $y_i^{(s)} = 0$ is a delinquent borrower. That is, p_1 and p_2 are the conditional probabilities that loans will be paid back in the two data generating regimes.)

For both data sets, $s = 1, 2$, do the following:

a) Fit a *logistic regression model* $\hat{p}_s^{\log}\colon \mathbb{R}^3 \to (0,1)$ on the *training data* $(x_i, y_i^{(s)})$, $i = 1, \dots, m$. Calculate the cross-entropy loss of $\hat{p}_s^{\log}$ on the training and test data. You can use the function sklearn.linear_model.LogisticRegression for this.

b) For SVM classification, we denote by $\hat{\sigma}_j$ the empirical standard deviation of $(x_{ij})_{i=1}^m$ and work with the normalized data $\tilde{x}_{ij} = x_{ij}/\hat{\sigma}_j$ (for both training *and* evaluation).

(i) Fit a SVM $\hat{f}_s^{\mathrm{svm}}\colon \mathbb{R}^3 \to \mathbb{R}$ of the form

$$\hat{f}_s^{\mathrm{svm}}(x) = \langle w, \Phi(x) \rangle + b$$

with feature map Φ on the *training data* using the hinge loss, kernel $k(x, x') = \exp\big(-\frac{1}{10}\|x - x'\|_2^2\big)$ and regularization parameter $\lambda = \frac{5}{2m}$. You can use the function sklearn.svm.SVC for this (the given choice of λ corresponds to the parameter $C = 1/(2\lambda m) = 0.2$ in sklearn.svm.SVC).

(ii) On top of $\hat{f}_s^{\mathrm{svm}}$, fit a *logistic function* $\hat{g}_s : \mathbb{R} \to (0,1)$ of the form

$$\hat{g}_s(z) = \frac{1}{1 + \exp(\alpha z + \beta)} \quad \text{for parameters } \alpha, \beta \in \mathbb{R}$$

so that $\hat{p}_s^{\mathrm{svm}} := \hat{g}_s \circ \hat{f}_s^{\mathrm{svm}}$ predicts conditional probabilities that loans are paid back; see Platt (1999)[1]. To this end, you may simply use the option `probability=True` in the sklearn.svm.SVC function.

(iii) Compute the cross-entropy loss of $\hat{p}_s^{\mathrm{svm}}$, $s = 1, 2$, on both the normalized training and test data.

(iv) Would the results change if we used standardized data $\tilde{z}_{ij} = (x_{ij} - \hat{\mu}_j)/\hat{\sigma}_j$ instead of the normalized data $\tilde{x}_{ij} = x_{ij}/\hat{\sigma}_j$, with $\hat{\mu}_j$ the empirical mean of $(x_{ij})_{i=1}^m$? Explain why or why not.

c) Generate FDR/TPR-curves (with the TPR on the horizontal axis and the FDR on the vertical axis) from the test data for $\hat{p}_s^{\log}$ and $\hat{p}_s^{\mathrm{svm}}$, and compute the AUC of each of these curves. Briefly explain how you construct the curves, and briefly interpret what the AUC means in this setting. Compare the two models on both data sets.

3. Let us now focus on the second dataset $(x_i, y_i^{(2)})$, $i = 1, \dots, m+n$. The goal is to find "good investment opportunities" in the *test data set* based on the *features* x_i, $i = m+1, \dots, m+n$. We here assume that loans are either completely repaid with interest or fully delinquent. In reality, a lender tries to recover parts of delinquent loans.

We compare three different lending strategies:

(i) We give out a loan to every person in the test data set in the amount of CHF 1000 charging an interest rate of 5.5%.

(ii) We only charge an interest rate of 1%, but we selectively choose the applicants who are awarded a loan (in the amount of CHF 1000) using the selection criterion

$$\hat{p}_2^{\log}(x_i) \geq 95\%.$$

[1] `https://home.cs.colorado.edu/~mozer/Teaching/syllabi/6622/papers/Platt1999.pdf`

(iii) We only charge an interest rate of 1% but we selectively choose the applicants who are awarded a loan (in the amount of CHF 1000) using the selection criterion

$$\hat{p}_2^{\text{svm}}(\tilde{x}_i) \geq 95\%,$$

where $\tilde{x}_i = (\tilde{x}_{i1}, \tilde{x}_{i2}, \tilde{x}_{i3})$ are the normalized features from 2.b).

To estimate the performance of the strategies (i)–(iii) above, we simulate different market scenarios according to the conditional probabilities $p_2(x_i)$, $i = m+1, \ldots, m+n$. Using independent Unif$(0,1)$-distributed random variables $\xi_{i,k}$, $i = 1, \ldots, n$, $k = 1, \ldots, 50000$, generate the $n \times 50000$-matrix $D \in \{0,1\}^{n \times 50000}$ given by

$$D_{i,k} = \begin{cases} 1 & \text{if } \xi_{i,k} \leq p_2(x_{m+i}) \\ 0 & \text{otherwise,} \end{cases}$$

where $D_{i,k} = 1$ means that in scenario k, the i-th loan is paid back with interest. So, the k-th column of D describes which loans are paid back in the k-th scenario.

Now, for each of the strategies (i)–(iii) above ...

a) plot a histogram of the profits & losses over the different market scenarios and estimate the expected profit & loss.

b) estimate the 95%-VaR of the profit & loss distribution (= negative of the 5%-quantile).

Report all profits & losses and VaRs in CHF.

c) Compare the three strategies and briefly interpret your results.